

— ONE SENTENCE

We help factories find quality and machine problems earlier.

Industrial Predictive Quality watches images, sensors and process data together. It tells the team what looks unusual, when the process started drifting, and what to inspect first.

Visual inspection

Machine risk

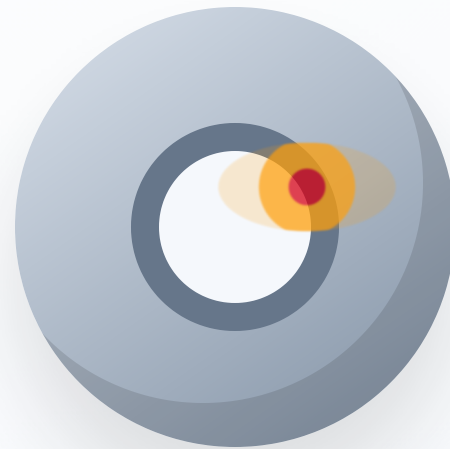
Process deviation

4-6 week pilot

— EXPLAIN IT LIKE I AM FIVE

Like a quality assistant that never stops watching.

A human expert knows what a good part, cycle or machine usually looks like. Our MVP learns those normal patterns from factory data and raises a clear review queue when something starts to deviate.



— WHY AUDI SHOULD CARE

Premium manufacturing loses money when small deviations become late defects.

Surface quality

Small marks, paint issues or assembly defects can escape manual checks.

Machine drift

Tools, robots, presses, conveyors and test benches can drift before they fail.

Too many signals

Cameras, PLC, SCADA, torque logs and test logs are often reviewed separately.

Late decisions

The cost rises when a defect reaches the next station, batch or customer.

It converts factory data into a ranked inspection list.

Collect

Images, sensors, process values, test logs



Learn normal

What good production usually looks like



Prioritise

Part, cycle, tool, batch or line risk

Output 1

Heatmap: where the visual anomaly is.

Output 2

Risk score: what to review first.

Output 3

Timeline: when the process started drifting.

— PRODUCT MODULES

Three practical products, one pilot.

1. Visual Quality

For parts, paint, surfaces, PCBs, packaging, welds or camera inspection. It produces heatmaps and a review queue.

2. Machine Risk

For vibration, current, temperature, torque, test logs or SCADA. It ranks machines/cycles by failure-soon risk.

3. Process Deviation

For setpoints, recipes and operating context. It compares expected evolution with actual behaviour.

— BUSINESS VALUE

The buyer does not buy AI. They buy fewer surprises.

QUALITY



Catch questionable parts earlier.

SCRAP



Reduce late rework and wasted material.

DOWNTIME



See process drift before breakdown.

ATTENTION

Top 10%

Focus experts where the risk is highest.

— CURRENT EVIDENCE

What we can honestly show today.

AREA	RESULT	BUSINESS MEANING	HOW TO PRESENT IT
Visual inspection	DINOv2/ResNet + PatchCore/PaDiM reached near-perfect quick-run scores on MVTec bottle.	The visual module is the closest to a product demo.	“Strong enough to pilot on real client images.”
Machine risk	Engineered sensor features are strong; learned future signal is promising but not a replacement.	Start with reliable physical indicators, then add learned signals if they improve validation.	“We benchmark honestly; we do not sell a black box.”
Hard regimes	Sensor signals help when lifecycle/cycle proxies are removed or when material/hardness changes.	Useful when simple counters stop explaining quality/risk.	“This is where the product becomes interesting.”

These are internal MVP results on public industrial datasets. They justify a pilot, not a production guarantee.

— HOW TO SAY IT AT AUDI

Use a simple factory story.

The story

“Imagine a line where every part image, every torque curve and every sensor trace becomes one ranked review queue. The system does not replace your quality engineers. It tells them where to look first.”

The ask

“Give us one process, one defect/risk objective and a small historical dataset. In 4-6 weeks we show whether this can reduce inspection burden, false alarms or late defect detection.”

Where to propose the first pilot.

Body & paint

Surface marks, paint defects, panel issues, weld spot anomalies, camera inspection review.

Assembly & torque

Torque curves, press-fit signals, robot stations, tool wear, repeat deviations by station.

Supplier quality

Incoming part inspection, batch ranking, visual defect heatmaps, supplier comparison.

Battery / e-drive

Test logs, thermal traces, end-of-line anomalies and early warning of process drift.

Maintenance

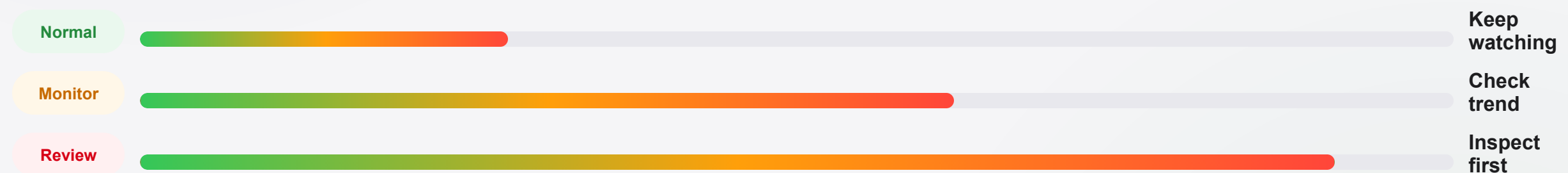
Vibration, current, temperature and SCADA for machine risk scoring.

Line management

Patch → part → cycle → batch → line hierarchy for practical alert dashboards.

— TRAFFIC-LIGHT STYLE WITHOUT GIMMICKS

Green means normal. Amber means monitor. Red means review first.



The point is not to show a fancy dashboard. The point is to reduce decision time: which part, station or machine needs human attention first?

— UNDER THE HOOD, IN PLAIN ENGLISH

We use proven building blocks before experimental research.

Production-minded baseline

For vision, use strong pretrained visual features plus memory/heatmap methods. For sensors, use engineered physical features plus calibrated risk models.

Research extension

Then test whether learned “expected future” signals add value. If they help, they become part of the pilot. If not, the client still receives a useful baseline system.

DINOv2 / ResNet

PatchCore / PaDiM

Engineered sensor features

Optional world-model surprise

— PILOT PLAN

4-6 weeks, one concrete line or process.

Week 1

Data audit: images, sensors, logs, labels, failure/quality events.

Week 2

Baselines: simple rules, visual foundation, sensor/process baseline.

Weeks 3-4

Modeling: heatmaps, risk scores, alert ranking, calibration.

Weeks 5-6

Review: metrics, examples, business case and integration decision.

— WHAT WE NEED

Ask for a small, concrete dataset.

Minimum data

100-1,000 images or historical sensor windows from one process, plus timestamps and any known quality/failure labels.

Best data

Images + sensor traces + process settings + line/station IDs + inspection result + maintenance or quality event history.

The first pilot should be narrow: one product family, one line, one quality/risk question.

— WHAT THE CLIENT RECEIVES

A decision package, not just a model.

Demo

Local HTML/PDF showing examples, heatmaps, scores and ranked alerts.

Metrics

AUPRC/AUROC, Precision@10, false alarm rate, lead time when applicable.

Go / no-go

Clear recommendation: integrate, refine, change data collection or stop.

— WHAT NOT TO SAY

Keep the pitch confident, but credible.

Say this

“We can run a fast pilot to prove whether your images, sensors and process logs can predict quality/risk earlier than current rules.”

Do not say this

“This is guaranteed to work in production immediately” or “we replace your engineers.” The offer is a rigorous pilot with practical outputs.

— MEETING SCRIPT

30-second opener.

“EVOCON already works in industrial automation and project support. This MVP adds an AI layer for predictive quality: it combines camera inspection, sensor trends and process context to produce a ranked review queue. We are not asking you to replace your systems. We propose a 4-6 week pilot on one line to measure whether we can catch defects or machine drift earlier, with heatmaps, risk scores and a go/no-go report.”

— FIRST QUESTION TO AUDI

“Where do you currently inspect too late?”

If they say visual defects

Propose visual pilot with heatmaps and review ranking.

If they say machine downtime

Propose sensor/process risk scoring with lead-time metrics.

If they say too many alarms

Propose alert prioritisation and false-alarm reduction.

— COMMERCIAL CLOSE

The safest ask is a pilot, not a platform sale.

Start with one process. Prove the value. If the metrics and expert review agree, extend to more lines, plants or suppliers.

Low risk

Uses existing data

Local demo

Clear go/no-go

— FINAL MESSAGE

We turn factory data into earlier warnings and clearer actions.

Industrial Predictive Quality is not a magic AI claim. It is a practical pilot: visual heatmaps, machine risk scores, process deviation timelines and a ranked inspection queue.

EVOCON Solutions

Pilot-ready MVP

Audi / automotive fit